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Are patients benefiting from participation in the German skin cancer screening programme? A large cohort study based on administrative data*

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Summary

Background The German programme for skin cancer screening was established in 2008 with the aim of reducing skin cancer mortality. However, the effectiveness and risk–benefit ratio of the programme remain unclear.

Objectives To compare the mortality rates of patients with melanoma who participate in a screening programme to those who do not.

Methods A retrospective cohort study, based on pseudonymized health insurance data of 1 431 327 individuals from Saxony, Germany, was conducted for the period 2010–2016. Patients with prevalent and incident melanoma were defined based on diagnosis, medical procedures and prescriptions. Patients who underwent screening and had a first diagnosis of melanoma within 2 years of screening were assigned to the intervention group. Relative survival and Cox regression were used to assess potential differences in mortality.

Results We identified 4552 individuals with prevalent and 2475 individuals with incident melanoma. The percentage of screening participants (n = 1801) who had locoregional (4·2% vs. 13·5%) and/or distant metastases (4·3% vs. 8·0%), or who were treated with systemic anticancer therapies (11·6% vs. 21·8%) was lower vs. nonparticipants (n = 674). Screening participants had significantly better survival rates. The unadjusted Cox model gave a hazard ratio (HR) of 0·37 [95% confidence interval (CI) 0·30–0·46]. After adjusting for named confounders, the effect remained (HR 0·62, 95% CI 0·48–0·80).

Conclusions Patients who participated in the screening programme had lower mortality than those who had not undergone screening. However, these findings may result from a healthy screen bias and/or overdiagnosis associated with screening, and not from the screening itself.

What is already known about this topic?

- There is limited evidence of the effectiveness of the German skin cancer screening programme and its associated risks more than 10 years after its initiation.
- The reduction in mortality as a result of early diagnoses that was hoped for has so far not been demonstrated.

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Conflicts of interest
Unrelated to this study, J.S. has acted as a consultant for Lilly, Novartis, Sanofi and ALK, and has received institutional funding from Sanofi, Pfizer, ALK and Novartis for investigator-initiated trials.

Data availability statement
The study was supported by AOK PLUS, with whom the Center for Evidence-based Healthcare has a Data Use and Transfer Agreement. The personal data of the beneficiaries were pseudonymized through AOK PLUS before data sharing. Personal identifiers were masked or deleted (exact names were pseudonymized prior to receiving the data; no social security numbers were provided). Quasi-identifiers were generalized (year of birth only, deletion of the last two digits of the zip code, etc.). The processing and analysis of sensitive data took place exclusively on the specially protected servers at Dresden University Hospital. Results of data analyses were shared in aggregate form only. The data are confidential and therefore not publicly available.

*Plain language summary available online

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What does this study add?

- A reduction in mortality is shown, but there is a high potential for bias.
- Secondary data alone cannot explain the facts of the matter.

The worldwide incidence of melanoma has been increasing rapidly over the last few decades. The reasons for this increase are controversial.[1–4] There is evidence that at least part of the increase cannot be explained by more examinations and better diagnostics alone.[5,6] Melanoma is a disease of elderly people. Therefore, some of the increase in incidence can be explained by the ageing population. Another reason for the increase may be overdiagnosis,[7] i.e. when diseases that may have never become a relevant health problem for the patient are detected and treated at an early stage. Unfortunately, being diagnosed with cancer can have harmful consequences for patients, for example through insecurity and discomfort due solely to the diagnosis.[8] Finally, a change in exposure to environmental risk factors, in particular intermittent intense sunbathing or indoor tanning, may also modify the personal risk of melanoma.[1,4,9]

Despite the increasing incidence of melanoma, melanoma mortality rates are generally low and have been stable worldwide over recent years, indicating that the current therapies are effective.[1] However, tumour stage at the initial diagnosis is a proxy for an individual's prognosis, with lower stages being associated with better outcomes.[10] In order to make progress in the early detection of skin tumours, the German programme for skin cancer screening was introduced in 2008.[11] The skin cancer screening is performed by specially trained primary care physicians and dermatologists. It is designed as an opportunistic screening, and there are no calls for participation. Individuals aged ≥ 35 years insured by statutory health insurance are eligible for a biannual asymptomatic screening, which is covered by the health insurance provider. Although the programme has been accompanied by an evaluation of its benefits, clear evidence on its effectiveness is still missing. The programme is therefore not without controversy.[12]

The aims of this study were to update the preliminary results on the use of the programme based on a large regional cohort from health insurance data, and to analyse potential differences in mortality for patients with melanoma with respect to participating or not participating in the screening programme.

Materials and methods

Study population

We used routine data from AOK PLUS, a large statutory health insurance in Saxony, Germany, which covers almost half of the regional population. The age distribution and sex ratio of AOK PLUS beneficiaries in Saxony are comparable to the Germany-wide population.[13] The data include information from inpatient and outpatient care with regard to diagnosis, procedures and prescriptions, as well as sociodemographic information (e.g. age and sex) of the insured population. A total of 1 431 327 million persons aged ≥ 35 years from the years 2010 to 2016 were available as the base on which all analyses were performed. Besides age in the year 2013, birth cohorts were formed in 10-year increments. Furthermore, we stratified patients by sex and selected comorbidities (Table 1). In order to incorporate a proxy for healthy users,[14] the proportion of persons who had received a voluntary vaccination against influenza was determined. We stratified patients who did and did not participate in screening by type of metastasis, and compared the proportions of individuals with systemic anticancer therapy.

Case definitions

In accordance with the good practice guidelines of the German Society of Epidemiology for secondary data analysis,[15] patients with prevalent melanoma were identified by at least one inpatient diagnosis [International Statistical Classification of Diseases and Related Health Problems, 10th Revision, German Modification (ICD-10-GM) C43] or by a combination of specific diagnoses and procedures in the outpatient sector (Table S1; see Supporting Information). Individuals had to be melanoma-free for at least 3 years (washout phase) to be recognized as newly diagnosed. Patients with a first onset of cancer could thus be estimated for the years 2013–2016. In addition, all patients had to be continuously insured during the period 2010–2012. Inspections of conspicuous lesions could be distinguished from an asymptomatic screening based on different billing codes. We placed patients with melanoma aged ≥ 35 years with a first diagnosis of melanoma within 2 years of screening into the intervention group with participation in the screening programme. The 2-year period was chosen owing to the biannual eligibility for an asymptomatic screening. Patients with incident melanoma and no participation in screening were assigned to the control group.

Incidence estimation

We used the European standard population to standardize our estimated cancer incidence rates per 100 000 persons annually to create comparability with other publications, particularly the region's clinical cancer registry. For comparison with external data from epidemiological cancer registries, we used the 1976 standard. However, we also included the standard from 2013, and reported on both.[16]

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Table 1 Baseline study information

	AOK PLUS population (2010–16)		Prevalent melanoma (2010–16)[a]		Patients with incident melanoma, 2013–16 (n = 2475)		AOK PLUS population			
					Screened[b]	Not screened[b]	Screened[b]		Not screened[b]	
Total no. of participants (age ≥ 35 years)	1 431 327 (100)		4552 (0·3)		1801 (72·8)[c]	674 (27·2)[c]	688 708 (48·1)		742 619 (51·9)	
Mean (SD) age in 2013 (years)	63·9 (15·3)		68·9 (13·2)		67·1 (12·5)	69·1 (13·1)	63·1 (14·2)		64·7 (16·2)	
Birth cohort										
1900–1909	645 (0·0)		0 (0·0)		0 (0·0)	0 (0·0)	70 (10·8)		575 (89·1)	
1910–1919	20 923 (1·5)		45 (0·2)		10 (0·6)	5 (0·7)	4401 (21·0)		16 522 (79·0)	
1920–1929	156 926 (11·0)		518 (0·3)		131 (2·3)	76 (11·3)	54 407 (34·7)		102 519 (65·3)	
1930–1939	266 966 (18·6)		1347 (0·5)		482 (26·8)	213 (31·6)	131 953 (49·4)		135 013 (50·6)	
1940–1949	238 942 (16·7)		1132 (0·5)		526 (29·2)	154 (22·8)	133 480 (55·9)		105 462 (44·1)	
1950–1959	305 325 (21·3)		773 (0·2)		327 (18·2)	115 (17·1)	156 077 (51·1)		149 248 (48·9)	
1960–1969	308 520 (21·5)		576 (0·2)		256 (14·2)	89 (13·2)	148 208 (48·0)		160 312 (52·0)	
1970–1975	133 080 (9·3)		161 (0·1)		69 (3·8)	22 (3·3)	60 112 (45·2)		72 968 (54·8)	
Sex										
Female	796 891 (55·7)		2157 (0·3)		830 (46·1)	314 (46·6)	395 780 (57·5)		401 111 (54·0)	
Male	634 436 (44·3)		2395 (0·4)		971 (53·9)	360 (53·4)	292 928 (42·5)		341 508 (46·0)	
Selected comorbidities										
CKD (N18)	232 824 (16·3)		1085 (23·8)		358 (19·9)	162 (24·0)	108 628 (15·8)		124 195 (16·7)	
Liver diseases (K71-76)	173 557 (12·1)		906 (19·9)		330 (18·3)	138 (20·5)	96 018 (13·9)		77 538 (10·4)	
CHD (I20-25)	339 795 (23·7)		1602 (35·2)		560 (31·1)	260 (38·6)	165 897 (24·1)		173 897 (23·4)	
HF (I50-52)	314 805 (22·0)		1348 (29·6)		454 (25·2)	207 (30·7)	144 523 (21·0)		170 281 (22·9)	
Hypertensive diseases (I10-15)	705 972 (49·3)		3352 (73·6)		1250 (69·4)	520 (77·1)	355 922 (51·7)		350 049 (47·1)	
T2DM (E11-14)	344 882 (24·1)		1606 (35·3)		568 (31·5)	256 (38·0)	164 293 (23·9)		180 588 (24·3)	
Procedures prior to inclusion (1 year)										
Influenza vaccination	813 172 (56·8)		3178 (69·8)		915 (50·8)	298 (44·2)	454 862 (66·1)		358 310 (48·2)	
Type of metastasis (after initial melanoma)										
None	–	–	–	–	1546 (85·8)	473 (70·2)	–	–	–	–
Locoregional within 100 days	–	–	–	–	75 (4·2)	91 (13·5)	–	–	–	–
Distant within 100 days	–	–	–	–	77 (4·3)	54 (8·0)	–	–	–	–
Locoregional after 100 days	–	–	–	–	43 (2·4)	20 (3·0)	–	–	–	–
Distant after 100 days	–	–	–	–	60 (3·3)	36 (5·3)	–	–	–	–
Therapy										
Systemic anticancer therapy within 30 days of initial melanoma diagnosis	–	–	–	–	209 (11·6)	147 (21·8)	–	–	–	–

Data are n (%) unless otherwise stated. CHD, coronary heart disease; CKD, chronic kidney disease; HF, heart failure; T2DM, type 2 diabetes mellitus.[a]Percentage share of total AOK PLUS cohort specified for each category; [b]percentage share in relation to the screening or nonscreening group; [c]percentage share among all patients with incident melanoma ≥ 35 years (n = 2475).

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Two different methods were used to calculate incidence rates from health insurance data. In the first approach, the years 2010–2012 were defined as a fixed melanoma-free period for the entire cohort, and – from 2013 onwards – cases were considered to be incident/newly diagnosed. In the second approach, the melanoma-free period was postponed by 1 year each time and was thus always 3 years for all persons. In the last case, it is, in principle, possible that people can be considered as an incident case several times, depending on the length of the period under review.

Statistical analysis

Based on the case definitions (Table S1), it was possible to estimate the total participation rate in the screening programme, as well as annual participation, in relation to all eligible individuals. As indicators for disease severity at the time of initial diagnosis, we used systemic anticancer therapies within 30 days, and locoregional or distant metastases documented within 100 days of the initial melanoma diagnosis. Comparative relative survival rates were calculated according to Perme and Pavlič.[17] The mortality of patients with melanoma was compared with the mortality of the general German population,[18] and also to the mortality of the Saxon population (received upon individual request to the State Statistical Office of the Federal State of Saxony). This enabled an estimation of excess disease-related mortality.

Two methods for the estimation of net survival were used (Ederer II and Pohar Perme). A maximum follow-up time of 4 years was considered. We estimated the number needed to screen (NNS) based on the observed difference in raw mortality between participants in screening and nonparticipants [absolute risk reduction (ARR)] and expanded it with a presumed 4-year prevalence of melanoma of 0·9/1000 persons for Saxony in 2015 according to the following formula: $NNS = (1/ARR) \times 100/(0{\cdot}9/1000)$. In addition, a proportional hazards model (Cox model) was used to estimate the effect of screening participation on survival.[19] The Cox model was adjusted for confounding patient- and disease-related factors. We selected age, sex, education level and 31 comorbidity groups, according to Elixhauser et al. (modified by Garland et al.),[20,21] as confounders. Influenza vaccination 1 year prior to diagnosis, a health check-up within the 2 years prior to diagnosis, as well as colorectal, prostate and breast cancer screening within the 3 years prior to initial diagnosis of melanoma, were used to adjust the Cox model for health-seeking behaviour. As model predictors, we also included a personal history of nonmelanoma skin cancer within the observation period prior to the diagnosis of melanoma, as well as the year of diagnosis and the implementation of a systemic therapy within 30 days of initial melanoma diagnosis. The Cox model was stratified into five groups according to the timing and type of metastases in relation to the initial melanoma diagnosis (none; locoregional within 100 days; distant within 100 days; locoregional > 100 days; distant > 100 days). Stratification was used rather than adjustment, to allow for different baseline hazards for these groups. Figure S1 (see Supporting Information) provides a rationale for stratifying baseline risk into five distinct groups. In addition, we recorded the individual occurrence of stroke (ICD-10 GM I61-64), ischaemic heart disease (I20-25) or severe heart failure (I50) in the final 3 months of observation as competing causes of death. Statistical analyses were performed in R.[22] We used the *epitools* package on all direct age standardizations, except for the calculation of relative survival and hazard ratios (HRs). For this, we used the *relsurv* and the *survival* packages.[23] All binomial proportion 95% confidence intervals (CIs) were calculated according to the Wald method.

Sensitivity analysis

We excluded all patients with a metastasis within 100 days of the initial diagnosis of melanoma, as they could no longer benefit from screening (early diagnosis), and ran the Cox regression again. Further, we tried to approach the problem of a possible length/lead time bias.[24] To this end, we discarded all patients with a survival time < 360 days from both groups (screening and nonscreening groups) to compensate for a presumed extension in lead time within screening participants. Because Breslow tumour thickness data were not available in our dataset, we included a recorded sentinel lymph node biopsy (SLNB) procedure as a predictor in the Cox model as a proxy for thinner tumours. In addition, we tried a second definition of screening participation. Individuals with a one-time occurrence of a screening code before the initial diagnosis of melanoma, regardless of the time interval, were included in the intervention group. The results were contrasted to the main analysis. We also shortened the washout phase from 3 to 2 years to extend the follow-up in another sensitivity analysis.

Compliance with guidelines and recommendations

This study adhered to the principles of the Declaration of Helsinki,[25] and to the principles of Good Epidemiological Practice and Good Practice in Secondary Data Analysis;[15,26] it also followed the recommendations for reporting observational studies.[26,27] The study was carried out in accordance with the General Data Protection Regulation of the European Union.

Ethics approval and consent to participate

An application to the Saxon State Ministry for Social Affairs and Consumer Protection to obtain consent for data transmission and analysis on behalf of the insured participants according to paragraph 75 SGB X (Zehntes Buch Sozialgesetzbuch - German federal law) was approved. The present study was a retrospective observational study based on statutory health insurance data.

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The results are presented in aggregated form only. Conclusions about individual persons were therefore not possible.

Results

In total, 688 708 eligible individuals in Germany were screened at least once between 2010 and 2016, constituting 48% of all eligible persons aged ≥ 35 years (n = 1 431 327). An average of 1 301 017 individuals were eligible for screening every year. The mean annual participation rate was 15·2% (mean n = 196 228), but participation increased steadily from 12·4% in 2010 to 17·6% in 2016.

We included 1 431 327 individuals of whom 4552 had prevalent melanoma between 2010 and 2016. After applying our incidence criteria (Table S1), 2475 individuals were identified with new-onset melanoma. Of these, 1801 (72·8%) participated in screening within a period of 2 years prior to their initial melanoma diagnosis, while 674 (27·2%) were diagnosed without screening (Table 2). Initial diagnoses of melanoma were found for 704 (28·4%) individuals within 30 days or less after screening. The mean age of the overall study population was 63·9 years; 55·7% of the population was female (Table 1). The mean age of patients with melanoma who participated in screening was 67 years; for patients who did not attend screening (control group), it was 69 years (Table S1). Men were somewhat more frequently affected. We found significant differences between participants in screening and nonparticipants in terms of immediate systemic anticancer therapy within 30 days (11·6% vs. 21·8%) and the occurrence of locoregional (4·2% vs. 13·5%) or distant metastases (4·3% vs. 8·0%) within 100 days of the initial diagnosis of melanoma (Table 1).

We report only incidences adjusted according to the European standard from 1976, to allow for comparability with cancer registry data from the region. There was little difference between the use of a fixed washout phase (3-year period without cancer), which was extended with each year of observation, or a dynamic washout phase, which was always only 3 years for each patient (see the subsection 'Incidence estimation'). Age-adjusted annual melanoma incidence rates per 100000 insured individuals based on all eligible patients with melanoma (n = 2475) within the AOK PLUS population ranged from 13·8 (95% CI 11·9–16·1) to 15·4 (95% CI 13·4–17·8) for females and from 18·9 (95% CI 16·7–21·4) to 21·6 (95% CI 19·2–24·4) for males, using the European standard from 1976 and a fixed washout phase. The annual average from the years 2014 and 2015 was 14·4 (95% CI 12·5–16·7) for females and 19·4 (95% CI 17·2–22·0) for males (Table 3). In sum, we observed an increase in melanoma incidence over time for Saxony. Estimates of standardized incidence ratios (SIRs) between screening participants and nonparticipants resulted in a mean SIR of 2·36 (95% CI 2·15–2·59) over the study period. Considering annual SIRs, no time trend could be observed.

The crude absolute mortality risk (Table 2) for individuals who participated in screening and who were diagnosed with melanoma (9·5%) was > 50% lower than that for individuals who did not take part in screening (22·8%). As we were interested in a hypothetical screening effect, we estimated a NNS to avoid one death among patients with melanoma during our observation period of 4 years (mean 660·5 days). The NNS was estimated to be 8321 (95% CI 7561–9249).

The relative survival of screening participants was considerably higher than for nonparticipants (Figure 1), regardless of the reference population (Germany, Saxony) and the fit method (Ederer II, Pohar Perme). We present a comparison with the German general population and the Ederer II fit method to achieve comparability with other publications. While survival of screening participants was comparable with that of the general German population, nonparticipants had significantly worse survival [survival ratio after 1 year 0·90 (95% CI 0·87–0·93); survival ratio after 3 years 0·82 (95% CI 0·77–0·88)].

The adjusted HR for screening participants with melanoma for dying within the observation period vs. nonparticipants was 0·62 (95% CI 0·48–0·80), indicating a reduced risk for screening participants. Univariate and adjusted HRs are provided in Table 4. Re-analysis of data excluding individuals with metastasis within 100 days of initial diagnosis (sensitivity analysis) showed a slightly more favourable adjusted HR (0·57; 95% CI 0·41–0·80) for screening participants vs. the main analysis. In addition to the definition of screening participation within 2 years prior to initial melanoma diagnosis, a second definition was contrasted here. All individuals with a single screening code prior to initial diagnosis were included. However, this only added 155 individuals to the screening group and the corresponding HRs remained stable. A lead time correction of 360 days did not qualitatively change the estimators. The HRs in the adjusted models were no longer significant (Table 4). Also, a shortening of the washout phase

Table 2 Raw numbers of individuals diagnosed with melanoma, with or developing a metastasis, and dying within the observation period (maximum 4 years)

Population total (n)	Subgroups	Screening	Share (%)	No screening	Share (%)
2475	With melanoma	1801	72·8	674	27·2
456	With metastases	255	14·2	201	29·8
325	Died with melanoma	171	9·5	154	22·8

The raw absolute risk reduction (ARR) in mortality for screening participants was estimated to be 13·3%.

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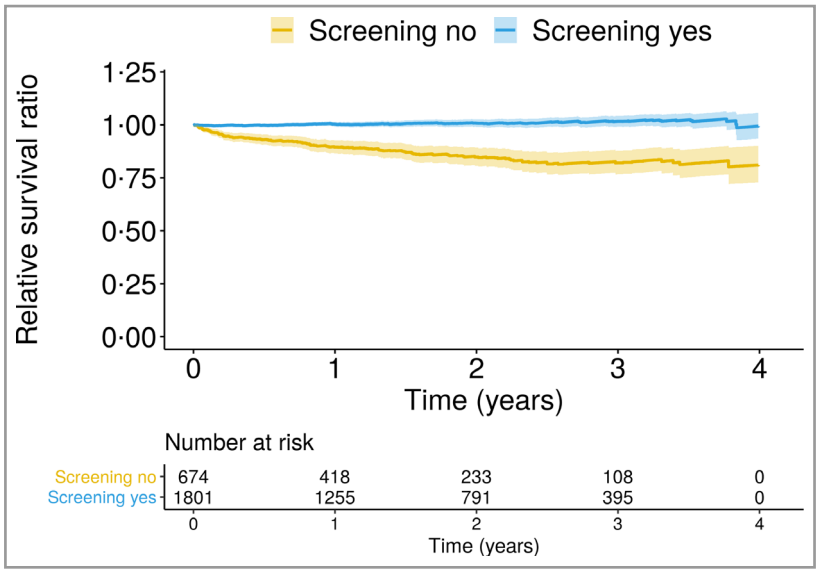


Figure 1 Relative survival of patients with newly diagnosed melanoma, stratified by skin cancer screening participation. Step functions of the survival curves are plotted with 95% confidence intervals. Censoring events are not shown. Risk table with absolute numbers of persons under risk is given beneath the graph. Screening participants had a better survival rate than nonparticipants over the entire follow-up period of a maximum of 4 years.

Table 3 Estimates of melanoma incidence by sex and year for Saxony based on AOK PLUS data

			Yearly incidence rates per 100 000 males			Yearly incidence rates per 100 000 females		
Cancer-free period		Year	Crude rate	Adjusted rate	95% CI (adjusted rate)	Crude rate	Adjusted rate	95% CI (adjusted rate)
3 years fixed	2013 standard	2013	33·84	29·23	26·08–32·82	25·25	19·44	17·10–22·11
		2014	33·46	27·99	24·97–31·39	23·75	18·39	16·09–21·00
		2015	34·62	29·70	26·54–33·22	26·27	20·20	17·79–22·94
		2016	36·21	30·91	27·68–34·51	28·02	21·13	18·67–23·94
		2014–15 average	34·04	28·85	25·76–32·31	25·01	19·30	16·94–21·97
3 years dynamic		2013	33·84	29·23	26·08–32·82	25·25	19·44	17·10–22·11
		2014	34·38	28·75	25·69–32·19	24·11	19·28	16·89–21·99
		2015	35·53	30·46	27·26–34·02	27·07	20·82	18·38–23·60
		2016	37·23	31·78	28·50–35·42	29·00	21·82	19·32–24·66
		2014–15 average	34·96	29·61	26·48–33·11	25·59	20·05	17·64–22·80
Cancer-free period								
3 years fixed	1976 standard	2013	33·84	19·79	17·54–22·40	25·25	14·54	12·61–16·86
		2014	33·46	18·89	16·72–21·40	23·75	13·79	11·89–16·08
		2015	34·62	19·94	17·68–22·58	26·27	14·96	13·00–17·29
		2016	36·21	21·59	19·18–24·37	28·02	15·40	13·42–17·76
		2014–15 average	34·04	19·42	17·20–21·99	25·01	14·38	12·45–16·69
3 years dynamic		2013	33·84	19·79	17·54–22·40	25·25	14·54	12·61–16·86
		2014	34·38	19·35	17·16–21·88	24·11	14·66	12·67–17·03
		2015	35·53	20·49	18·19–23·16	27·07	15·43	13·44–17·79
		2016	37·23	22·20	19·76–25·02	29·00	15·89	13·88–18·27
		2014–15 average	34·96	19·92	17·68–22·52	25·59	15·05	13·06–17·41
GKR publication[a]		2014–15 average	26·4	16·1	–	19·2	12·2	–

Estimates were based on all patients aged ≥ 35 years with a newly diagnosed melanoma between 2013 and 2016 (n = 2475). Crude rates were standardized using the European standard population (1976, 2013) and are reported per 100 000 person-years. Ninety-five per cent confidence intervals (CI) for the adjusted rates are given. [a]Averaged sex-specific rates of incident melanoma in Saxony according to Gemeinsamen Krebsregister der Länder (GKR) Berlin (BE), Brandenburg (BB), Mecklenburg-Vorpommern (MV), Sachsen-Anhalt (SN) und der Freistaaten Sachsen (ST) und Thüringen (TH) – Schriftenreihe 4/2018.

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Table 4 Survival analysis based on Cox model

		Screening participation within 2 years before initial diagnosis (n = 1801)				At least one screening participation before initial diagnosis (n = 1956)			
Predictor (model type)	Reference	Total	n	Sensitivity[a]	n	Total	n	Sensitivity	n
Screening (univariate)	No screening	0·37[b]	2475	0·38[b]	2178	0·43[b]	2475	0·39[b]	2178
Screening (univariate)[c]	No screening	0·50[b]	1688	0·51[b]	1557	0·51[b]	1688	0·49[b]	1557
Screening (adjusted)[d]	No screening	0·62[b]	2459	0·57[b]	2165	0·68[b]	2459	0·48[b]	2165
Screening (adjusted)[c,d]	No screening	0·75	1680	0·76	1549	0·71	1680	0·67	1549

Corresponding hazard ratios (HRs) and number of evaluable persons for each comparison are given. Baseline HRs for all multivariate models were stratified into five groups. [a]Patients with metastasis within 100 days after initial melanoma diagnosis were excluded. [b]$P < 0·05$. [c]Lead time correction (360 days were subtracted from both groups). [d]Adjusted for age group (three); sex; year of diagnosis; education; systemic anticancer therapy; history of nonmelanoma skin cancer; influenza vaccination in the year before the initial melanoma diagnosis; colorectal, prostate and breast cancer screening within 3 years prior to diagnosis; health check-up (from 35 years of age onwards) within the 2 years prior to diagnosis; 22 Elixhauser comorbidities; stroke, ischaemic heart disease and heart failure in last 3 months; five strata for type/timing of metastasis.

to 2 years and the associated extension of the follow-up by 156 days, on average, did not change the results. The HR in the unadjusted Cox model was 0·41 (95% CI 0·34–0·48). Consideration of a performed SLNB procedure was highly significant (HR 0·51, 95% CI 0·35–0·75) but had no relevant effect on the mortality risk difference between screening participants and nonparticipants.

Discussion

The results presented here suggest a beneficial effect of the German skin cancer screening programme on prognostic factors for patients with a melanoma diagnosis. Patients who participated in screening had lower Union International Cancer Control (UICC)/American Joint Committee on Cancer (AJCC) stages at time of diagnosis, approximated by the lower proportion of locoregional or distant metastases within 100 days of initial diagnosis, compared with nonparticipants. Further, the proportion of patients who received systemic anticancer therapy within 30 days of the initial melanoma diagnosis was lower in screening participants than in nonparticipants. Because melanoma can become dangerous even at a very small size, patients who present conspicuous lesions to the doctor are more likely to be suffering from more advanced stages of melanoma. Thus, the screening programme seems to result in a benefit. It leads to the detection of malignant melanoma at earlier stages. This selection alone could drive the difference in mortality between screening participants and nonparticipants, as screening participants may show no symptoms and have no history of melanoma. The general health condition of screening participants may be better than that of nonparticipants. This selection effect, which occurs particularly in screening studies, is called 'healthy screenee bias',[28,29] and cannot be excluded in our study, owing to its observational nature.

Furthermore, our data did not allow for the detection of cases that are prone to overdiagnosis and would hence have a better prognosis.[30,31] Melanomas at low tumour stages are usually resected without the need of further therapy. Thus, we do not have any distinction criterion at hand to measure the proportion of overdiagnoses in German health insurance data for patients with melanoma. If the share of overdiagnoses were higher in screening participants than in nonparticipants, this would result in a mortality reduction in screening participants.[8]

Unlike previous studies,[10,12,32,33] our observations of the associations of the programme with survival and mortality rates are strong. Patients who participated in screening had a significantly higher rate of survival and reduced mortality. These results remained statistically stable after controlling for various covariates. Unfortunately, only a follow-up period of 4 years could be analysed for survival estimates. Therefore, besides the 'healthy screenee bias' or overdiagnosis related to screening, another possibility for reduced excess mortality in screening participants could be the detection of disease at earlier stages. Thus, the percentage of patients who survive longer after their initial diagnosis is naturally larger than that of a cohort who first present with more advanced stages (length/lead time bias). Whether the 1-year period we used is appropriate for correction is uncertain. According to our analysis, the beneficial effect of screening on survival was maintained over the 4-year observation period. The NNS for this time period was estimated to be 8321. We are aware that this estimate is hypothetical, as any selection bias and overdiagnosis related to screening was ignored, and the entire effect was attributed to the screening itself. Based on evidence from clinical trials, the NNS for Haemoccult® screening to prevent one cancer-specific death within 8·5 years was estimated to be 808. For mammograms, the NNS was reported to be 695 in women aged 60–69 years within 9 years of follow-up.[34] Our estimate for skin cancer screening exceeds these numbers by a factor of 10. During the study period, numerous new therapies for malignant melanoma were approved,[35,36] including BRAF in combination with mitogen-activated protein kinase inhibitors, or immune checkpoint inhibitors,[37] which could

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also affect melanoma-specific mortality. Altogether, despite strong associations, we were unable to separate the screening effect from potential interferences ('healthy screenee bias', overdiagnosis, lead time bias and availability of new therapies). More evidence is needed.

The observed increase in melanoma over time is in accordance with other studies from Germany and elsewhere in Europe.[10,38–40] The incidence of melanoma estimated in this analysis for the years 2014/15 by using the European standard from 1976 was 14·4 cases per 100 000 person-years for women and 19·4 cases for men. Using cancer registry data from Saxony, these annual average values were 12·2 for women and 16·1 for men for the same time period. Cancer registry data from Germany provided values of 19·9 for women and 21·0 for men for the year 2016, putting our estimates in the same range as those based on cancer registry data. In addition, there could be an overestimation of our actual case numbers due to (i) an erroneous inclusion of false-positive melanoma cases by our case definitions and/or (ii) a possible mixing of prevalent and incident cases, especially at the start of the study period. However, there could also be a slight under-recording in the cancer registry data from Saxony due to unreported cases. A further limitation is that no direct information on the tumour stage was available in the health insurance data and the severity of the tumour was determined solely by the status of metastases.

In general, citizen awareness of the programme has grown, which is evidenced by the increasing annual participation rates. The impact of the programme on the number of diagnoses is also high, as almost two-thirds of all melanoma diagnosis were made in screening participants. Whether the increasing incidence trend is mainly due to improved diagnostic techniques and associated down-shifts in thresholds for the characterization of melanoma, leading to overdiagnosis related to screening, could not be determined here.[8]

Our results showed a positive association between the prognostic parameters of patients with melanoma and participation in the general skin cancer screening programme in Germany. Screening participants were diagnosed with lower UICC/AJCC stages and therefore had fewer radical treatments in advance than patients diagnosed without having participated in screening. Owing to a relatively short observation period, the long-term effects of the programme could not be analysed adequately. The observed differences in survival times within the first years of diagnosis may have been caused by a mixture of selection bias, and overdiagnosis or lead time bias in relation to screening and not by the screening itself.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Figure S1 Survival probabilities (Kaplan–Meier estimates) of different strata within patients newly diagnosed with melanoma.

Table S1 Case definitions.

Powerpoint S1 Journal Club Slide Set.

Video S1 Author video.

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